

High-Resolution Entropy Theory:

Defining Time from the Irreversible Energy Dissipation in Quantum Networks

-Section 2 and Section 3 of 'Multiplicative-Entropy-Driven Invariant-Topology Two-Layer Chiral-Space (MITC) Model'
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Abstract:

This chapter introduces a novel theoretical framework for understanding the nature of time, based on the concept of discrete Space Elementary Quanta (SEQ) that constitute the fabric of spacetime. In this model, time is not an independent background parameter but emerges from the transformation or state update processes of the SEQ network. Each transformation corresponds to a unique entropy value, defined as the product of the energy norms of the participating SEQs. This multiplicative entropy encodes the progression of time, making it inherently discrete and finite due to the limited number of possible transformations in a finite universe.

The chapter emphasizes that time is operationally defined through the counting of spatial transformations, aligning with a thermodynamic perspective where entropy increase dictates the arrow of time. This approach establishes a direct link between time evolution and the second law of thermodynamics, ensuring a natural emergence of causality from the irreversible nature of entropy growth.

Furthermore, the chapter outlines how this time framework is consistent with both quantum mechanics and relativity. Local time is introduced as a measurable quantity tied to the transformation count within a specific spatial region, which corresponds to the concept of proper time in special relativity. The differences in time perception across reference frames arise from the cumulative variations in SEQ transformation rates, offering a foundational explanation for relativistic time dilation effects.

In the following sections, the paper will elaborate on the mapping between time and multiplicative entropy, exploring how entropy encodes causality and ensures a directional flow of time. Additionally, the mechanisms underlying time dilation in both special and general relativity—such as how spatial deformation affects the SEQ network's transformation frequency—will be discussed in detail. These developments aim to provide a unified description of time that bridges thermodynamics, quantum field theory, and gravitational physics.

Introduction:

What is time? How do we understand time? How do we define time? These are frequent questions on Q&A platform, and they are also questions that philosophers, physicists, scholars from interdisciplinary fields, and ordinary people have repeatedly contemplated and pursued. It can be said that understanding time is a fundamental philosophical and physical question.

Throughout history and across cultures, philosophers and physicists have offered various perspectives on the nature of time:

Subjective theory of time: Some philosophers emphasize that time is merely the human consciousness perceiving changes in the external world. From Parmenides, Kant, to Husserl, these thinkers have highlighted the relationship between time and human consciousness.

Objective theory of time: Some philosophers and physicists emphasize that time is a count of motion or change processes. **Time-as-background theory:** This view considers time as an independent variable. Some modern physicists hold this view, as did Newton and others in classical physics. In modern quantum field theory, time is similarly treated as a background parameter, although some exploratory branches of quantum field theory consider time as a relational variable within a quantum system.

Time as the process of change in things: This understanding follows the objective theory of time. Modern thinkers such as Leibniz, Whitehead, and contemporary causal set theory all regard time as a collection of events and their relationships.

Spacetime unification: General relativity introduced the concept of spacetime as a unified whole, emphasizing the co-evolution of space and time. General relativity also introduced the idea that the metric field modulates local proper time.

Programming assigns time parameters to events, marking the sequence of events.

Irreversibility of time: Modern thermodynamic studies have revealed that change in real-world phenomena is at least macroscopically irreversible—this is known as the thermodynamic arrow of time or the law of entropy increase. Considering the event-set view of time, the law of entropy increase provides a fundamental driving force behind causal relationships between events.

Locality and non-locality of time and space: Both quantum field theory and general relativity traditionally treat time and space as local, limited by the speed of light, with event relationships following spatial and temporal adjacency. However, some contemporary theories, such as quantum entanglement theory, hypothesize the existence of non-local interactions, supported by a series of experiments.

The following are my views:

The Full Text of Preprint: The Nature of Time, The Essence of Gravity, The Structure of Space, The Formula of Entropy, The Factors of Causality’ <https://doi.org/10.5281/zenodo.14788393>

a. In fact, all theories of time share a common point: time is a measure or count of change—it is a scale or count of change, whether in a continuous or discrete view of change, whether subjective or objective. Subjective time is human perception of change, while objective time is the count of change in the universe as a whole or in its parts.

b. Only a theory of time that incorporates the thermodynamic arrow of time can be considered a good philosophical or physical understanding of time, because the law of entropy increase has been empirically verified, and no counterexamples have been found yet, whether at the microscopic or macroscopic level. Even at the QED level, even a single electron (there is no truly stationary electron) generates entropy through radiation.

c. From the perspective of the rate of change, general relativity and related observational experiments have shown that gravitational fields indeed modulate the frequency of change in local regions—time slows down. That is to say, time is not an independent variable but a relational variable dependent on spatial change. When space is distorted or disturbed by energy, the rate of change in the corresponding local region also changes.

d. The evolution of things and the locality of space and time in their interactions form the foundation of contemporary physics and are also supported by empirical observations. Here, I want to discuss the issue of non-locality: if experiments like those violating Bell's inequality at low energy levels lead to non-local interactions in spacetime, then the laws of physics would not be what we currently observe. Moreover, the object of physical investigation would be an extremely complex system with a state space larger than the computational capacity of the universe itself. From this perspective, non-locality is not valid.

My view: My perspective is close to that of Einstein, Leibniz, Whitehead, and contemporary causal set theory. I believe that time is the count of cosmic change, the refreshment of cosmic states, the set of events, and the set of transformations. In fact, this understanding is easily accepted by professionals in the computer science field. At the same time, time is not an independent variable but is influenced by the change in space itself, as has been verified by observations and experiments, such as gravitational time dilation and gravitational redshift. Furthermore, I believe that space is quantized, and time is the set of spatial transformations. A transformation set of discrete sets is still a discrete set. To fully understand time, we must also distinguish between local time and cosmic global time. The time dilation in special relativity actually reflects different observers in different motion frames counting changes differently, rather than a concept of objective proper time. The gravitational time dilation in general relativity, however, describes how spatial distortion affects the local rate of change.

The following is the outline of my article on the nature of time. By constructing a multiplicative entropy-time-space transformation set with multiple mappings, I attempt to embed the thermodynamic arrow of time—its irreversible nature of energy distribution uniformization—into both general relativity and quantum field theory.

2. Time as a counting process of spacetime network transformations:

2.1 SEQ serve as the electromagnetic wave conducting medium. Matter with mass and its motion are waves in this medium. In this framework, nothing truly moves through space - light speed c is the maximum conduction speed c , preventing velocity stacking beyond c . All physical phenomena correspond to specific energy state configurations, establishing SEQ as the universal substrate.

2.2 The universe's composition: Energy conservation and quantization imply a finite number N of SEQ, each with M possible energy states, (where each energy state m_i is an integer multiple of Planck's constant h ,) allowing up to M^N transformations. These M energy states form an algebraic system incorporating translational, spinning and rotational operations connecting to standard model. Energy conservation and entropy increase constraints reduce possible transformations significantly below M^N .

2.3 Time definition:

2.3.1 Let J be the possible universe transformations ($J \ll M^N$).

2.3.2 The Planck time (t_p) interval separates adjacent transformations as the minimal time unit.

2.3.3 Time's arrow follows entropy increase.

2.3.4 Transformations map non-bijectionally to entropy values (k distinct values partition J transformations into K classes). Parallel transformations share the same entropy values, but only one can occur. The entropy set maps to possible time values - each moment corresponds to one universe transformation. Non-uniform entropy increase means only a subset of possible time values actually occur.

2.3.5 Each space transformation (state transition of the SEQ network) can be assigned a unique entropy value calculated via the multiplicative energy distribution across this space transformation's matrix.

2.3.6 Finite transformations ensure discrete, limited time in this model.

(Note: this derivation is a speculative exploration within the discrete framework and does not imply reality.)

Subsequent chapters will introduce a high-resolution, computable definition of multiplicative entropy and the mapping between multiplicative entropy and time; the mechanisms of time dilation will also be discussed. Readers are welcome to continue exploring these topics in the following sections.

3. Definition and formula of Multiplicative Analytic Entropy

In this definition of entropy, the entropy value of a closed system at a given moment (i.e., during a specific state transition) is calculated as the multiplicative product of the energy norms of all SEQ involved in that transition (that moment's space transformation).

Entropy value of closed system $S = \prod_{i=1}^n m_i, i \in N$ ①

Energy of closed system = constant = $\sum_{i=1}^n m_i, i \in N$ ②

$S_{\max} \leq m_i^n$, When all m_i are equal or differ only by Planck's constant h

(Where m_i refers to the energy carried by the i th SEQ during a single transformation of the closed system, where each energy state m_i is an integer multiple of Planck's constant h , $m_i=n_ih$, $n_i \in N$)

Each SEQ carries energy m_i = the resonant kinetic energy of this SEQ+its spin kinetic energy+the elastic potential energy of the local space surrounding the SEQ, calculated based on the gradient of resonance frequency shift relative to the Planck frequency of the SEQ, as will be elaborated in Section 9.1 and Section13.

Under this setup, the total energy of the entire space can indeed be represented solely by m_i , without the need to separately calculate the elastic potential energy within the sub-Planckian components.

3.1. Energy transfer rules and triggering conditions:
 Energy exchange occurs between adjacent SEQ (i,j) if and only if the following thermodynamic gradient exists: $m_i > m_j + h$, Energy transfer occurs only in discrete quanta of Planck's constant h , $m_i \rightarrow m_i -h$; $m_j \rightarrow m_j +h$ (Planck's constant: h)

3.2. Numerical Example: System States and Entropy Evolution

Table.1: Simplified Entropy Increase Demonstration

System State	SEQ Energy Distribution $\sum_{i=1}^n m_i=12$	Entropy $\prod_{i=1}^n m_i$	Remarks
Initial non-equilibrium state	[3, 1, 5, 3]	45	-
Intermediate state	PathA[3, 1, 4, 4]; PathB[3, 2, 4, 3];	A48; B72;	-
Final state	PathA[3, 2, 3, 4]; PathB[3, 3, 3, 3];	A72; B81;	Due to adjacent energy transfer with minimal quanta h , this system cannot reach maximum entropy in case A

Note:The above analysis demonstrates that different entropy-increasing pathways exhibit distinct sequences of entropy variation.

3.3. Logarithmic Relation:
 After logarithmic transformation, $\ln S$ aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.4. Proof of Spontaneous Entropy Increase
Spontaneity Theorem of entropy increase (Second Law of Thermodynamics):
 For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be $m_i=a$, $m_j=b$ ($a>b+h$);

The post-transfer entropy ratio is:

$$S_{t2}/S_{t1}=(a-h)(b+h)/ab=1+h(a-b-h)/ab>1 \text{ (Planck's constant } h)$$

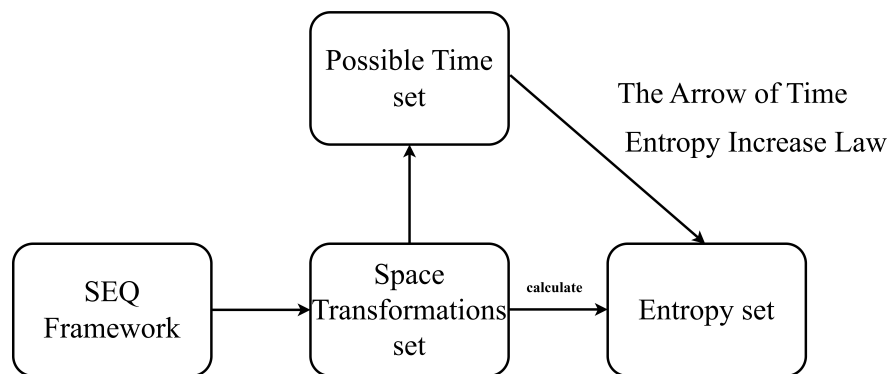
3.5. The definition conserves energy, has an entropy ceiling, ensures spontaneous increase, and logarithmically aligns with classical entropy.

Indeed, this model establishes a novel analytic quantum thermodynamic framework where coarse-graining automatically enforces the indistinguishability of **quantized homogeneous invariant network** in physical properties while simultaneously characterizing the entropy-increasing process of energy homogenization. These properties collectively confirm the validity of this entropy definition.

At this point, a clear multiple mapping can be established:

One space transformation→one entropy value→One possible moment

Space-Time-Entropy Mapping Diagram



Draw tool: draw.io [Computer software]. <https://github.com/jgraph/drawio>

Diagram.1: Space-Time-Entropy Mapping Diagram

This time-entropy mapping realizes the analytical form of Planck's entropy concept within a discrete spacetime network and naturally embeds the thermodynamic arrow of time into the ADM formalism of General Relativity. In this framework, temporal evolution emerges as a statistical ordering of energy redistributions in SEQ network, unifying quantum thermodynamics and spacetime geometry under a single discrete dynamical structure.

3.6. The Spontaneous Entropy Increase is Causality.

The existence of isentropic transformations implies that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, a certain dynamic freedom persists.

The existence of isentropic transformations and different entropy-increasing

pathways imply that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, with a certain dynamic freedom persists.

3.7. Why Analytic Entropy is Adopted: Within this model's framework, physical states at Planck-time scales exhibit deterministic characteristics. Although current experimental conditions cannot directly measure them, traditional statistical entropy can be reduced to analytic expressions at this scale. This perspective not only reveals the microscopic essence of statistical quantities but also provides a new analytic foundation for quantum thermodynamics—unifying macroscopic statistical behaviors with deterministic dynamics at the Planck scale.

Multiplicative analytic entropy corresponds to each step of energy homogenization in a system. With every step toward a more uniform energy distribution, the multiplicative analytic entropy increases, offering a clearer and more detailed characterization of the process compared to traditional statistical entropy.

Multiplicative Entropy vs. Traditional Statistical Entropy

Comparison Dimension	Multiplicative Entropy	Traditional Statistical Entropy
Process Explicitness	Explicitly records energy homogenization steps via product sequences (e.g., $\prod_i m_i$), preserving microstate transition details	Describes only macro-state differences via logarithmic state-count ($\ln \Omega$), erasing intermediate dynamics
Physical Intuitiveness	Entropy increase directly reflects irreversible energy redistribution; time asymmetry emerges from dynamics	Relies on probabilistic assumptions (e.g., molecular disorder) and requires ad hoc low-entropy past boundary
Process Resolution	Tracks Planck-timescale (t_p) energy transfers;	Limited to ensemble averages, incapable of resolving quantum fluctuations or short-timescale entropy production

3.8. Analysis of the Maximum Entropy Principle

In the SEQ model, the maximum entropy principle is manifested through the driving tendency of entropy increase: energy not only flows from a higher-energy SEQ to an adjacent lower-energy SEQ, but it also follows the path with the largest energy difference.

For example, consider an SEQ i , which is adjacent to two other SEQ j and k . The energies carried by nodes i , j , and k are A , B , and C respectively, where $A > B > C$. In this case, there are two possible energy transfer paths from SEQ i according to the principle of entropy increase: path $i \rightarrow j$ or path $i \rightarrow k$. We will analyze the entropy change for each path separately.

Under the constraint of energy conservation (i.e., $A + B + C = \text{constant}$), the entropy of the local system composed of these three SEQ before energy transfer is: $S = A \times B \times C$

Path $i \rightarrow j$: transferring energy to node j (which has relatively higher energy):

$$S' = (A - 1)(B + 1)C = (A - 1)(BC + C)$$

Path $i \rightarrow k$: transferring energy to node k (which has lower energy):

$$S'' = (A - 1)B(C + 1) = (A - 1)(BC + B)$$

Here, "1" represents one unit of Planck constant h . Since $B > C \Rightarrow BC + B > BC + C \Rightarrow S'' > S'$, the entropy increases more along the $i \rightarrow k$ path—that is, the path with the larger energy difference leads to a greater increase in entropy.

This deduction can be easily generalized to cases where the number of adjacent nodes is greater than two, so a detailed proof is omitted here.

However, it should be emphasized that the path with the maximum energy difference is not necessarily unique. Therefore, for states that have not yet occurred, the future still retains sufficient degrees of freedom — the evolution is not entirely deterministic.

3.9. In section 9.1.1, we will continue to discuss the decomposition of energy carried by SEQ $\{m_i = K_{\text{resonant}} (K_{\text{spin}}) + U_{\text{elastic}}$ assigned to this SEQ from its adjacent elastic bonds manifested as frequency suppression.), the relational expressions, and the supplementary description of the initial low-entropy state of the universe. Chapter 13 will provide a detailed explanation of the metric-SEQ resonance frequency mirror relationship under this model. With this setting, there is no need to observe the sub-Planckian constituent-level elastic potential energy or the metric tensor; instead, the regional elastic potential energy can be directly represented by the SEQ resonance frequency. Chapter 14 will mention that SEQ spin and resonance frequency mutually excite each other and change synchronously; therefore, SEQ spin kinetic energy can also be represented by SEQ resonant kinetic energy.

3.10. The multiplicative analytic entropy proposed by this model, taking the Planck constant—the fundamental unit of energy quantization discovered by Planck—as its basic element and grounded in the dynamical evolution of a discrete spacetime structure, precisely responds to Planck's original pursuit of an analytic expression for entropy. It also provides a novel microscopic foundation for characterizing thermodynamic irreversibility and the pathway of energy homogenization, thereby emerging as a candidate framework for constructing an intrinsic thermodynamic arrow of time.